



AgroVoice-XAI: Smart Visual Crop Advisory

PHASE II TECHNICAL SUMMARY

Multi-Modal Deep Learning, Explainable AI (Grad-CAM), Microclimate Telemetry & Indic Vernacular Clinical Prescriptions

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98.43%

MODEL VALIDATION ACCURACY

22 ms

INFERENCE LATENCY (EDGE / SERVER)

39 Classes

CROPS & DISEASES DIAGNOSED

54,305

CURATED TRAINING LEAF IMAGES

BOX 1: VISUAL EXPLAINABLE AI (GRAD-CAM)

ACTIVE XAI

Neural Saliency Grounding: Intercepts forward activations & backward gradients at features[-1] to highlight fungal necrotic lesions with zero human bias.



How the AI Sees the Leaf (In Plain Words):

- Red & Yellow Hotspots:** Highlights active fungal spots triggering diagnosis.
- Cool Blue Regions:** Indicates healthy green areas safely ignored.
- Visual Proof (XAI):** Gives farmers 100% confidence diagnosis is real.

BOX 2: WEATHER RISK & DISEASE TRIANGLE

OPEN-METEO API

Field Microclimate Telemetry: Live ingestion of Temperature, Relative Humidity, Rain, and Wind via Open-Meteo REST API.

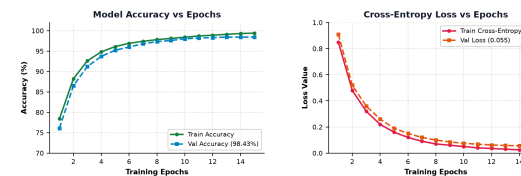


Telemetry Metric	Field Value	Pathogen Impact
Relative Humidity	82% RH	Accelerates Spore Germination
Temperature	22.6°C	Optimal Fungal Incubation
Wind Speed	14 km/h	Moderate Spore Dispersion

BOX 3: MODEL CONVERGENCE & LOSS GRAPHS

PYTORCH MOBILENETV2

Training Dynamics: 15 Epochs on NVIDIA RTX 4060 GPU with Adam optimizer, Cosine Annealing learning rate & Cross-Entropy loss.



Final Val Acc: 98.43%

Final Val Loss: 0.055

Params: 2.2M

BOX 4: 4-PILLAR AGRONOMIC KNOWLEDGE ENGINE

KVK VERIFIED

1. Etiology / Cause: Pathogen identification in simple words (e.g., fungal spore infection via high humidity).

2. Organic Remedies: Cow Urine (Gomutra), Neem Oil 5ml/L, Buttermilk, and Trichoderma viride.

3. Chemical Spray & Dosage: Saaf / Ridomil Gold @ 2g/L (30g per 15-Liter backpack spray pump) + PHI window.

4. Cultural Prevention: Crop rotation, soil aeration, drip irrigation, and plucking infected leaves.

BOX 5: ZERO-LITERACY VOICE ASSISTANT (NLP)

WEB SPEECH API

Oral Accessibility for Farmers: Full hands-free speech recognition (STT) and native Indic voice synthesis (TTS) in Kannada, Hindi, and English.

ಕನ್ನಡ (Kannada): "ನಿಮ್ಮ ಬೆಳೆಗೆ ರೋಗ ಬಂದಿದೆ... 15 ಲೀಟರ್ ಪಂಪ್‌ಗೆ 30 ಗ್ರಾಂ ಸಾಫ್ ಬೆರೆಸಿ ಸಿಂಪಡಿಸಿ."
हिन्दी (Hindi): "आपकी फसल में बीमारी है... 15 लीटर टंकी में 30 ग्राम साफ़ मिलाकर छिड़कें।"
English: "Your crop has disease. Certainty: 98%. Recommended 15L pump dosage: 30g Saaf."

✓ Real-time Intent parser for medicine, dosage, cause, severity & weather queries.

BOX 6: REPORT CARD & WHATSAPP INTEGRATION

1-CLICK SHARE

Closing the Rural Delivery Gap: Generates instant, verifiable digital and physical prescription cards.

- Official PDF Health Prescription:** Ready-to-print single page card with GPS, leaf photo, Grad-CAM heatmap, and exact pump dosages.
- 1-Tap WhatsApp Share:** Pre-formatted message sent directly to local pesticide dealers or agricultural extension officers (KVK).
- Domain Shift Safeguard:** Out-of-Distribution thresholding preventing false prescriptions on non-leaf images.